SURJECTIVITY OF AN ETALE MORPHISM OF MCQUILLAN AFFINE FORMAL ALGEBRAIC SPACES AND FAITHFUL FLATNESS OF THE RING MAP

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one identification of a spectrum with a fibre product; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The situation of the lemma identifying etale morphisms with completions of etale algebras, and the correspondence between morphisms and ring maps. Same chapter, Lemma lemma-etale, referred to in the statement at lines 4491-4492. Fixes the correspondence between the morphism f and the ring map B to the completion of A.
- An ideal consisting of topologically nilpotent elements lies in the Jacobson radical, and faithful flatness of a flat map can be tested modulo an ideal in the radical. Algebra, Lemma algebra-lemma-contained-in-radical at proof lines 4506-4508 and Algebra, Lemma algebra-lemma-ff-rings at proof lines 4508-4510. Gives the equivalence of the three ring-theoretic formulations.
- Morphisms from a quasi-compact scheme into the target factor through a quotient of the algebra. Same chapter, Lemma lemma-factor-through-thickening, invoked at proof lines 4517-4519. Reduces surjectivity of the morphism to the quotient maps.

Source reference. Original lemma 0AME: sources/stacks-chow/tags.txt, source lines 13402–13402. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00HQ: sources/stacks-chow/tags.txt, source lines 652–652. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN8: sources/stacks-algebra/pinned-original-source.tex, source lines 39507–39507. Pointer outside the counted span, invoked

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once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2080–2080. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *With notation and assumptions as in Lemma 0AN8 let $f : X \to Y$ correspond to $B \to A^\wedge$. The following are equivalent*

(1) *$f : X \to Y$ is surjective,*
(2) *$B \to A$ is faithfully flat,*
(3) *for every weak ideal of definition $J \subset B$ the ring map $B/J \to A/JA$ is faithfully flat, and*
(4) *for some weak ideal of definition $J \subset B$ the ring map $B/J \to A/JA$ is faithfully flat.*

Proof. Let $J \subset B$ be a weak ideal of definition. As every element of J is topologically nilpotent, we see that every element of $1 + J$ is a unit. It follows that J is contained in the Jacobson radical of B (Algebra, Lemma 0AME). Hence a flat ring map $B \to A$ is faithfully flat if and only if $B/J \to A/JA$ is faithfully flat (Algebra, Lemma 00HQ). In this way we see that (2) – (4) are equivalent. If (1) holds, then for every weak ideal of definition $J \subset B$ the morphism $\mathrm{Spec}(A/JA) = \mathrm{Spec}(B/J) \times_Y X \to \mathrm{Spec}(B/J)$ is surjective which implies (3). Conversely, assume (3). A morphism $T \to Y$ with T quasi-compact factors through $\mathrm{Spec}(B/J)$ for some ideal of definition J of B (Lemma 0AIA). Hence $X \times_Y T = \mathrm{Spec}(A/JA) \times_{\mathrm{Spec}(B/J)} T \to T$ is surjective as a base change of the surjective morphism $\mathrm{Spec}(A/JA) \to \mathrm{Spec}(B/J)$. Thus (1) holds. □